

What Happens When Purchase Intentions Are Blocked: Evidence from a Large Coffee Chain*

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Abstract

We study how temporary supply disruptions shape consumer demand and product exploration in a digital coffee retail setting. Leveraging detailed panel data on purchases and product assortments across 260 stores and 800,000 consumers, we exploit store closures during the Covid-19 pandemic as exogenous shocks to purchasing opportunities. A triple-differences design uses predicted purchase intention to isolate consumers for whom closures are likely to bind. While overall demand recovers quickly after reopening, treated high-intention consumers exhibit a sharp behavioral shift as they become less likely to explore new products. We rationalize this pattern with a goal-based mechanism. When an active purchase intention is interrupted, the underlying goal persists and intensifies, leading consumers to prioritize goal completion upon return. Heterogeneity by consumers' pre-disruption novelty-seeking further supports this interpretation: blocked intentions widen the gap between consumers who were already more versus less inclined to try new products. This mechanism explains why supply disruptions have limited effects on purchase frequency but significant, heterogeneous effects on product choice. Our findings highlight the role of goal persistence and consumer heterogeneity in shaping demand under constraints and show that supply-side disruptions can have lasting effects on product discovery despite rapid demand recovery.

Keywords: Customer Engagement, Customer Reactivation and Retention, Variety Seeking, Goal-Directed Consumption

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1 Introduction

Consumers often intend to purchase but are temporarily unable to do so due to supply constraints such as stock-outs or store closures. The natural benchmark is that such disruptions primarily shift demand over time, with consumers delaying purchases and resuming their usual behavior once constraints are lifted. Temporary supply shocks therefore affect the timing of purchases but not the underlying decision process. We study when this benchmark fails by distinguishing between consumers whose purchase intention is interrupted and those who simply would not have purchased during the disruption. When a closure blocks an active purchase intention, the interruption may affect not only when consumers buy, but also how they choose once they return.

This distinction raises two central questions. On one hand, do consumers whose intended purchases are blocked reduce subsequent purchase frequency, consistent with weakened demand, or do they return to their prior activity? On the other hand, does the interruption affect product choice? Specifically, do treated high-intention consumers change their propensity to explore new products relative to comparable low-intention consumers? These questions determine whether temporary constraints merely reallocate demand across time or also reshape consumer choice.

We address these questions in the context of a large-scale digital coffee retail platform, where we observe detailed panel data on individual purchases of 770,00 consumers, store-level product assortments and store operations across a network of 260 locations for 18 months. A key feature of the setting is the occurrence of temporary store closures during Covid-19 pandemic that interrupt consumers' ability to transact at specific locations. These closures are driven by external operational constraints that affect store availability over time and are not targeted to individual consumers. As a result, they generate plausibly exogenous variation in purchasing opportunities at the store level, while leaving the broader demand environment largely unchanged. Because closures affect all potential customers of a given store simultaneously, they create sharp, localized disruptions that allow us to track how the same consumers adjust their behavior once constraints are lifted.

A central challenge in this research setting is that the closure does not bind for all consumers. The disruption only affects those who would have purchased at the closed store during the closure period. To address this challenge, we use pre-disruption purchase behavior to construct a measure of ex-ante purchase intent at the store level. Consumers who frequently purchase from a given store prior to the disruption are more likely to have intended to purchase during the closure window. We exploit this variation in intent to implement a triple-differences design that compares (i) treated versus control consumers, (ii) before versus after the disruption, and (iii) high-intent versus low-intent consumers. This approach isolates the causal effect of displacement, which is the effect of preventing a consumer with an active purchase intention from completing a transaction, from the average effects of the disruption across the broader population.

Our analysis yields three main findings. First, temporary disruptions have limited effects on aggregate demand. Purchase frequency declines mechanically during the closure but recovers quickly after reopening, indicating that these shocks do not generate persistent reductions in consumption. Second, the average effects on product choice suggest an increase in variety seeking after reopening. However,

this pattern masks substantial heterogeneity across consumers. Treated high-intention consumers exhibit a qualitatively different response because they become significantly less likely to purchase new-to-them products after the disruption. The main displacement estimate is a 4.1 percentage point decline in novelty-seeking, about 8.3% of the pre-period mean, and only emerges when conditioning on ex-ante purchase intent.

We interpret these findings through a goal-based mechanism. When an active purchase intention is blocked, the underlying goal remains active and may intensify during the disruption. Upon reopening, this leads consumers to prioritize goal completion, directing purchases toward familiar or previously chosen products. We test this mechanism by comparing consumers with different pre-disruption novelty-seeking propensities. The estimates show that blocked intentions widen the novelty gap across consumer types, consistent with the idea that the interruption makes familiar consumption paths especially salient for consumers who were less exploratory to begin with.

Our paper contributes to the marketing literature along three dimensions. First, we study how temporary constraints interact with consumer intent to shape demand, highlighting a form of state dependence driven by persistent purchase goals rather than solely by past choices or habits. Second, we contribute to the literature on supply-side shocks by showing that their effects depend on whether the constraint binds for the consumer, emphasizing the importance of heterogeneity in exposure to disruptions. The closest applied benchmark is [Levine and Hristakeva \(2025\)](#), who study a temporary grocery strike and use predicted trip displacement to distinguish baseline demand changes from the effect of an interrupted shopping trip. We use the same basic logic of separating exposure from binding interruption, but shift the object of interest from store choice and trip counts to product exploration within a changing digital assortment. Third, we speak to the literature on product discovery by showing how temporary interruptions can change exploration even when demand levels recover quickly. More broadly, our results suggest that understanding demand under temporary constraints requires going beyond average responses and explicitly accounting for how disruptions interact with consumer intent. This perspective provides a framework for analyzing environments in which purchasing opportunities are intermittently constrained, and highlights the role of consumer goals in shaping choice.

2 Behavioral Mechanism and Micro-Foundation

2.1 Counterfactual Thinking and Goal Persistence

A growing literature in marketing and consumer behavior studies how consumers respond when intended purchases cannot be completed due to temporary constraints such as stockouts, unavailable products, or interruptions in access. The standard benchmark is that these disruptions primarily shift purchases across time, with consumers delaying consumption and resuming their prior behavior once constraints are lifted. Under this view, temporary supply-side disruptions affect the timing of purchases but do not fundamentally alter the underlying decision process. However, prior research in consumer psychology suggests that interrupted actions can generate persistent cognitive and motivational effects that extend beyond simple postponement.

Our framework builds on the literature on counterfactual thinking and goal persistence. Counterfactual thinking refers to the mental simulation of how events could have unfolded differently under alternative circumstances (Epstude & Roese, 2008). Importantly, this literature argues that counterfactual thoughts are not merely retrospective evaluations, but functional cognitive processes that shape future behavior by maintaining attention on unrealized goals and missed actions. When individuals are prevented from completing an intended action, they naturally construct upward counterfactuals such as “I would have purchased if the store had remained open.” These mental simulations increase the salience of the interrupted goal and motivate subsequent goal-completion behavior.

This perspective is particularly relevant in our setting. Temporary store closures do not simply eliminate purchasing opportunities at random. For consumers with an active purchase intention, the closure interrupts an ongoing consumption objective. As a result, the disruption creates a counterfactual consumption episode in which consumers mentally represent the purchase they would have made absent the closure. We argue that this counterfactual process generates a form of goal persistence that shapes post-reopening behavior.

We argue that counterfactual thinking represents the cognitive mechanism through which interrupted purchase intentions remain behaviorally active. When consumers are prevented from completing an intended purchase, they mentally simulate the forgone consumption episode. This counterfactual representation maintains the original consumption goal in an active state, leading consumers to return after reopening with more focused and less exploratory purchase behavior.

Our mechanism is closely related to research on interrupted goals and implemental mindsets in consumer behavior. Prior work shows that unfulfilled goals remain cognitively active and continue to influence subsequent decisions until they are completed (e.g. Gollwitzer, 1990; Kruglanski, Shah, et al., 2002; Masicampo & Baumeister, 2011). Similarly, research on shopping momentum demonstrates that consumers who enter an implemental mindset become more focused on goal completion and less deliberative in their subsequent choices (Dhar, Huber, et al., 2007). In our context, consumers whose purchases are blocked by store closures may remain in a suspended implemental state during the disruption period. Once stores reopen, these consumers return with a more focused purchase objective, prioritizing completion of the interrupted consumption goal rather than exploring alternative products.

This mechanism also relates to the literature on stockouts and constrained choice. Fitzsimons (2000) show that consumers’ responses to unavailable products depend critically on the strength of their commitment to the intended purchase. When commitment is high, constraints generate stronger behavioral reactions and alter subsequent decision-making. Our setting extends this insight by considering temporary store closures that interrupt not only access to a specific product, but the entire execution of a planned consumption episode. Rather than weakening demand, these interruptions may preserve or even intensify purchase intentions through counterfactual processing.

A recent cognitive theory of reasoning and choice provides useful language for this channel. Bordalo, Gennaioli, et al. (2026) argue that decision makers first recognize the kind of problem they face using contextual cues and past experiences, and that this categorization determines which attributes receive attention. Applied to our setting, a closure can change the post-reopening problem from ordinary exploration in a coffee assortment to completion of an interrupted purchase routine. The mechanism therefore

does not require consumers' tastes or information to change. Instead, the same consumer may place more attention on familiar products because the closure makes the familiar consumption path the natural category through which the next purchase is represented.

The implications of this mechanism are central to our empirical findings. If consumers whose purchase intention is blocked engage in counterfactual thinking and maintain an active purchase goal, then closures should not substantially reduce subsequent purchase frequency. Instead, consumers should return once constraints are lifted and complete the intended consumption activity. At the same time, because post-reopening behavior is driven by goal completion, consumers should become less likely to deviate from the originally intended purchase path. Consequently, consumers whose purchase intention is blocked should exhibit lower product exploration and reduced novelty-seeking after reopening.

2.2 A Behavioral Micro-Foundation

Consider a consumer who visits a coffee store and chooses between familiar products and new products. Familiar products are items the consumer has purchased before, while new products are items the consumer has not yet tried. In a standard choice framework, the probability of choosing a new product depends on intrinsic preferences, variety-seeking motives, and idiosyncratic taste shocks.

Our key departure is that a temporary closure can alter the consumer's attentional state. For consumers with an active purchase intention, the closure does not simply remove a transaction opportunity. It interrupts an intended consumption episode. This interruption induces counterfactual thinking: the consumer mentally represents the purchase she would have made had the store remained open. The counterfactual does not necessarily make one specific product more attractive. Instead, it reactivates the consumer's familiar consumption routine.

This distinction is important. In a coffee setting, the interrupted goal is often not "buy product j " in a narrow sense, but rather "complete my usual coffee purchase." The relevant object of salience is therefore the familiar consumption path: the set of products, routines, and choices that the consumer has previously experienced. Counterfactual thinking increases the cognitive accessibility of this familiar path. When the store reopens, the consumer returns with a more focused consumption objective, placing greater attentional weight on familiar products and less relative attention on unfamiliar alternatives.

The model can be summarized in three steps.

First, before the closure, the consumer has a familiar consumption routine formed through past purchases. This routine defines a set of familiar products that are cognitively accessible and easy to choose. New products may still be attractive, but they require attention, discovery, and willingness to deviate from the routine.

Second, the closure blocks the consumer's intended purchase. For consumers who would have purchased during the closure period, this generates a counterfactual representation: "I would have bought my usual coffee if the store had been open." This representation keeps the familiar routine active during the disruption.

Third, after reopening, the consumer makes a product choice under distorted attention. Familiar products receive an additional salience weight because they are linked to the interrupted consumption goal. New products do not receive this salience boost because they were not part of the counterfactual

consumption script. As a result, the probability of choosing a familiar product rises, and the probability of novelty-seeking falls.

The framework does not impose a signed prediction on purchase frequency after reopening. Because the blocked intention may lead to delayed goal completion, substitution, or discouragement, the purchase-incidence effect is an empirical object. The formal predictions instead concern novelty-seeking. First, at the population level, blocked intention should reduce novelty-seeking because counterfactual thinking increases the salience of old products. Second, this reduction should be stronger for low-novelty consumers, widening the novelty gap between high- and low-novelty types.

This framework clarifies the behavioral distinction between counterfactual thinking and goal-focused behavior. Counterfactual thinking is the cognitive process generated by the interruption. It makes the unrealized consumption episode mentally accessible. Goal-focused behavior is the observable consequence: consumers return to complete the familiar consumption routine, reducing exploration.

It also distinguishes the counterfactual-goal mechanism from the missed-exposure mechanism. Under the counterfactual-goal mechanism, consumers are aware of alternatives but pay less attention to them because the familiar routine is salient. Under the missed-exposure mechanism, consumers fail to explore because new products are less likely to enter their consideration set. Both mechanisms predict lower novelty-seeking, but they operate through different primitives: salience and motivation in the first case, information and awareness in the second.

Mapping to notation. The discussion above implies a simple modeling restriction. When a consumer's intended coffee purchase is blocked, the relevant counterfactual is not an arbitrary product draw. The consumer thinks about something she has purchased in the past: the old product, routine, or familiar consumption path that would have been completed if the store had remained open. We therefore remove the notation that assigns probabilities to new-versus-old counterfactual content and assume that counterfactual thinking works through the persistent salience of old products. The model begins with one consumer type and then introduces heterogeneity in the strength of this old-product salience.

2.2.1 Setup

Consider a consumer facing closure event e and post-reopening period t . Let

$$C_e \in \{0,1\} \tag{2.1}$$

indicate whether the consumer's focal store is closed during event e . Closure exposure can affect all treated consumers, regardless of whether they had an active purchase intention during the closure. Let

$$I_e \in \{0,1\} \tag{2.2}$$

indicate whether the consumer had a purchase intention during the closure window, meaning that she would have purchased from the focal store absent the closure. The blocked-intention state is

$$B_e = C_e I_e. \tag{2.3}$$

Thus, C_e captures baseline closure exposure, while B_e captures the incremental case in which an intended purchase is prevented from being completed.

Let $P_{et} \in \{0,1\}$ indicate whether the consumer purchases after reopening. Purchase probability is

$$\Pr(P_{et} = 1) = \Lambda \left(\alpha_{et}^P + \beta_P C_e + \kappa B_e \right), \quad (2.4)$$

where $\Lambda(\cdot)$ is the logistic CDF. The parameter β_P captures the baseline closure-exposure effect on purchase incidence, while κ captures the incremental effect of blocked intention. The model does not sign either parameter. If blocked consumers return to complete the intended purchase, then κ may be positive. If the counterfactual memory vanishes or offsetting forces dominate, then κ may be zero. If consumers substitute away or become discouraged by the interruption, then κ may be negative. Purchase incidence is therefore an empirical object rather than a formal prediction of the model.

2.2.2 One consumer type

Let $A_{et}^O \in \{0,1\}$ indicate whether at least one old product enters the consumer's post-reopening consideration set. Old products are products the consumer purchased before the closure. The key carryover parameter is $\lambda \geq 0$. It measures whether the counterfactual representation of the forgone purchase remains accessible after reopening. If $\lambda = 0$, the counterfactual memory trace vanishes and blocked intention has no post-reopening salience effect. If $\lambda > 0$, the old product or old routine represented during closure becomes more accessible after reopening. This formulation follows memory-based consideration models in which accessibility affects consideration and choice without necessarily changing primitive product evaluations (Nedungadi, 1990).

Old-product consideration is given by

$$\Pr(A_{et}^O = 1) = \Lambda \left(\alpha_{et}^O + \beta_O C_e + \lambda B_e \right), \quad (2.5)$$

where β_O captures the baseline effect of closure exposure on old-product consideration and is unrestricted. The blocked-intention effect on the log-odds of old-product consideration is therefore

$$\Delta^O \equiv \frac{\partial}{\partial B_e} \log \left(\frac{\Pr(A_{et}^O = 1)}{1 - \Pr(A_{et}^O = 1)} \right) = \lambda. \quad (2.6)$$

This replaces the earlier formulation in which the effect depended on the probability that the counterfactual was about a new product. Here, blocked intention increases the log-odds of old-product consideration directly because the counterfactual is assumed to be about something purchased in the past.

To connect consideration to observed novelty-seeking, let $N_{et} \in \{0,1\}$ indicate whether the purchased product is new to the consumer, conditional on purchase. After consideration is formed, product utilities are stable:

$$u_{jt} = v_{jt} + \varepsilon_{jt}. \quad (2.7)$$

Closure exposure and blocked intention do not enter v_{jt} . They affect observed choice by changing which

products are cognitively accessible. In particular, because old and new products compete for limited attention and choice, greater old-product salience reduces the relative probability that a new product is chosen. A compact logit representation is

$$\Pr(N_{et} = 1 \mid P_{et} = 1) = \frac{\sum_{j \in \mathcal{N}_t} \exp(v_{jt})}{\sum_{k \in \mathcal{O}_t} \exp(v_{kt} + \lambda B_e) + \sum_{j \in \mathcal{N}_t} \exp(v_{jt})}, \quad (2.8)$$

where $\mathcal{O}_t$ is the set of old products and $\mathcal{N}_t$ is the set of products new to the consumer. Equation (2.8) is not meant to impose that utilities change. Instead, the salience term summarizes the increased accessibility of old products in the post-reopening choice process. It implies

$$\frac{\partial}{\partial B_e} \Pr(N_{et} = 1 \mid P_{et} = 1) < 0 \quad \text{if } \lambda > 0. \quad (2.9)$$

Prediction 1 (Population-level reduction in novelty-seeking). If counterfactual salience persists after re-opening, $\lambda > 0$, blocked intention raises the log-odds of old-product consideration and reduces the probability of choosing a new product conditional on purchase. Therefore, in the empirical triple-differences design, the population-level displacement effect on novelty-seeking is negative:

$$\delta_N^D < 0. \quad (2.10)$$

If $\lambda = 0$, the counterfactual memory trace does not carry over, and the model predicts no blocked-intention effect on novelty-seeking through this channel.

2.2.3 Two consumer types

Now allow consumers to differ in intrinsic novelty seeking. Let

$$\tau \in \{H, L\}, \quad (2.11)$$

where H denotes high novelty-seeking consumers and L denotes low novelty-seeking consumers. Let v^τ denote the consumer's baseline tendency to consider and choose new products, with

$$v^H > v^L, \quad (2.12)$$

consistent with the view that novelty seeking is an individual-difference tendency that predicts exploratory consumption and product switching (Van Trijp, Hoyer, et al., 1996; Baumgartner & Steenkamp, 1996).

The strength of old-product salience is also type-specific. Low-novelty consumers are more likely to think about old products and routines when a purchase is blocked, so

$$\Delta^{O,L} = \lambda^L > \Delta^{O,H} = \lambda^H \geq 0. \quad (2.13)$$

Equivalently, blocked intention raises the log-odds of old-product consideration more for low-novelty

consumers than for high-novelty consumers. A type-specific version of old-product consideration is

$$\Pr(A_{et}^O = 1 \mid \tau) = \Lambda \left(\alpha_{et}^O + \beta_O C_e - v^\tau + \lambda^\tau B_e \right). \quad (2.14)$$

The term $-v^\tau$ captures that high-novelty consumers are less oriented toward old products at baseline, while $\lambda^\tau B_e$ captures the type-specific increase in old-product salience induced by blocked intention.

For novelty-seeking, the same limited-attention logic implies the reduced-form log-odds representation

$$\log \left(\frac{\Pr(A_{et}^N = 1 \mid \tau)}{1 - \Pr(A_{et}^N = 1 \mid \tau)} \right) = \alpha_{et}^N + \beta_N C_e + v^\tau - \lambda^\tau B_e, \quad (2.15)$$

where A_{et}^N indicates that at least one new product enters the consumer's consideration set. The negative sign on $\lambda^\tau B_e$ reflects the crowd-out of new-product consideration by persistent old-product salience.

Without blocked intention, the high-low gap in the log-odds of new-product consideration is

$$G_0 = v^H - v^L. \quad (2.16)$$

With blocked intention, the gap becomes

$$G_1 = (v^H - \lambda^H) - (v^L - \lambda^L) \quad (2.17)$$

$$= (v^H - v^L) + (\lambda^L - \lambda^H). \quad (2.18)$$

Because $\lambda^L > \lambda^H$, blocked intention widens the novelty gap between high- and low-novelty consumers.

Prediction 2 (Blocked intention widens the novelty gap). If counterfactual salience persists and low-novelty consumers think more about old products, $\lambda^L > \lambda^H$, then blocked intention reduces novelty-seeking more strongly for low-novelty consumers than for high-novelty consumers. The high-low novelty gap increases by

$$G_1 - G_0 = \lambda^L - \lambda^H > 0. \quad (2.19)$$

In an empirical specification with a low-novelty interaction, the model predicts a negative incremental low-type effect,

$$\delta_{N \times Low}^D < 0, \quad (2.20)$$

because the old-product salience generated by blocked intention is stronger for low-novelty consumers. Equivalently, in a specification where low-novelty consumers are the omitted group, the incremental effect for high-novelty consumers should be positive.

3 Setting and Data

We study consumer behavior using transaction-level data from a large Chinese quick-service coffee chain, where purchases are conducted primarily through a mobile application. The data span June 2020 to December 2021 and include approximately 10.6 million transactions from 779,000 customers.

The platform operates a dense network of physical stores with 260 locations and offers a dynamic product assortment that varies across stores and over time. Product availability is highly heterogeneous since a given product appears in 169 stores on average over its lifecycle, but is available in only 59 stores on a given day. This substantial variation implies that consumers' exposure to products depends critically on where and when they interact with the platform. The data track individual purchase histories at a high frequency, including the timing of transactions, store choice and product-level decisions. Consumers repeatedly interact with a relatively small set of nearby stores, generating persistent patterns in both purchasing frequency and product choice. This structure enables us to construct measures of baseline demand, store dependence and product exploration at the individual level.

A central feature of the setting is the presence of temporary store closures, which occur at the store-day level and prevent all consumers from transacting at the affected location. These closures arise from external operational constraints associated with Covid-19-related restrictions and generate sharp disruptions in purchasing opportunities. Because closures are not targeted to individual consumers and affect all potential customers of a store simultaneously, they provide plausibly exogenous variation in access to the platform.

Specifically, we focus on 18 unexpected store closures and construct a matched treatment-control design. Treatment customers are defined as those whose preferred store, identified based on pre-closure purchasing behavior, is the store that closes. Control customers are defined as those whose preferred store is one of a set of matched stores that remain open. The matching procedure ensures comparability across treated and control stores. For each closure, we select up to five control stores with similar store set-up times and exclude stores that are already used as controls for earlier closures. We then assign customers to treatment or control based on their preferred pre-closure store. To guarantee that store assignment is meaningful, we restrict attention to customers with at least five pre-closure purchase days and a preferred-store purchase share of at least 80%. The resulting panel contains 40,148 member-event pairs and 321,184 member-event-period observations for purchase frequency, including 7,390 treated and 32,758 control member-events. The purchase-intention classifier identifies 2,041 treated member-events, or 27.6% of treated episodes, as high-intention episodes whose intended purchases would be blocked by the closure. For novelty-seeking, the estimation sample counts 99,644 observations as the outcome is defined only for periods in which a purchase occurs. This setting generates rich identifying variation along three dimensions. First, we observe the same consumers before and after a disruption, allowing for within-consumer comparisons. Second, closures vary across stores and over time, providing cross-sectional and temporal variation in exposure to constraints. Third, consumers differ in their reliance on specific stores, creating heterogeneity in the extent to which a closure constitutes a binding constraint.

Figure 1 presents the raw closure shock for the two primary outcomes. The left panel plots purchase frequency, while the right panel plots novelty-seeking. Relative period 0 is the closure window. Purchase frequency falls sharply for treated customers during this window, confirming that closures create a binding local supply constraint. The novelty-seeking panel does not show a comparable drop because the outcome is defined conditional on making a purchase. The period-0 point therefore describes the composition of the purchases that still occur during the closure window, not the probability of being

Table 1: Treatment-control balance before closure

Variable	Control mean	Treated mean	Difference	SE
Member-events	32,758	7,390		
Unique members	32,758	7,390		
Predicted purchase-intention probability	0.307	0.313	0.006	0.004
High-intention share	0.269	0.276	0.007	0.006
Pre-period purchase frequency	0.050	0.046	-0.003***	0.001
Pre-period novelty-seeking	0.540	0.558	0.018***	0.006
Closure duration in days	16.9	18.2	1.3***	0.1

Notes: The unit is a member-closure event. Pre-period purchase frequency is averaged over relative periods -4 to -1. Pre-period novelty-seeking is averaged over pre-period windows in which the member purchased at least one product. Difference is treatment minus control. Standard errors are heteroskedasticity-robust difference-in-means standard errors at the member-event level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

able to purchase. After reopening, purchase frequency returns close to the control path, whereas the post-reopening composition of purchases remains the main object of the analysis below.

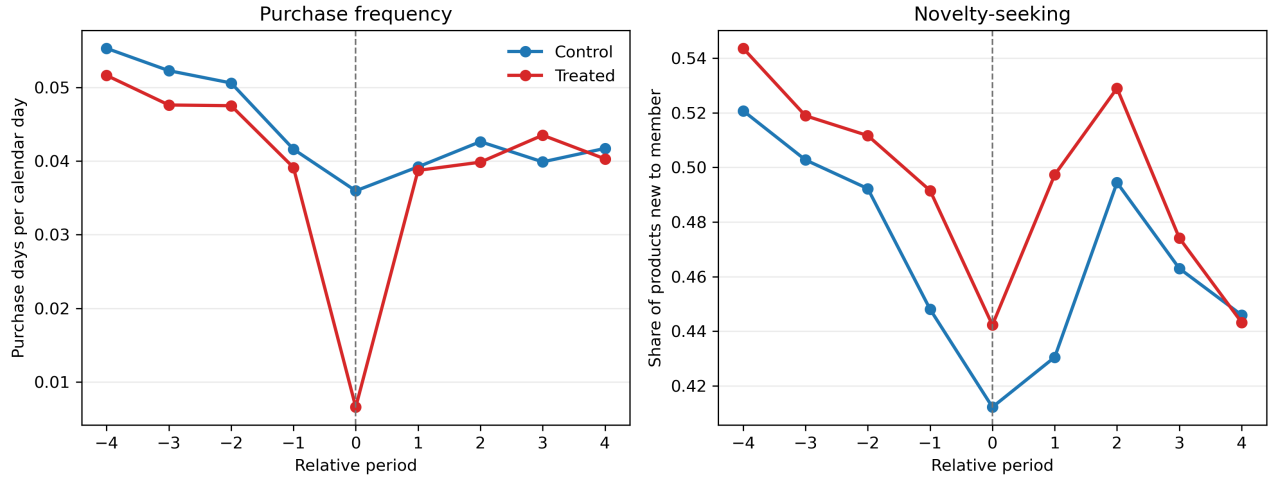


Figure 1: Raw closure shock in purchase frequency and novelty-seeking

Notes: The figure plots treatment and control means by relative event period for the 18-closure analysis sample. Relative period 0 is the closure window. Purchase frequency is purchase days per calendar day. Novelty-seeking is the share of purchased products that are new to the member and is observed only in member-periods with purchases.

Furthermore, the combination of repeated interactions, localized store dependence and time-varying product assortments creates an environment in which temporary disruptions can affect both the level and composition of demand. In particular, closures restrict purchasing opportunities and interrupt ongoing purchase processes. This structure allows us to study how temporary supply constraints affect consumer behavior along two margins: purchase frequency and product exploration. More broadly, the setting provides a natural laboratory to examine how consumers adjust when a planned purchase cannot be executed, and whether such interruptions have persistent effects on subsequent choices.

4 Empirical Strategy and Results

The central challenge in this setting is that not all treated customers are equally affected by a store closure. While all customers assigned to a treated store are exposed to the disruption, only a subset would have made a purchase during the closure window absent the interruption. Following the logic in [Levine and Hristakeva \(2025\)](#), we separate closure exposure from binding interruption by classifying member-events according to predicted purchase intention. A purchase is blocked only when a high-intention customer is also assigned to the treated store. Conceptually, the high-intention indicator captures more than purchase intensity. It serves as a revealed preference proxy for an active purchase goal. Repeated purchases from the chain reflect not only habit or inertia, but may also indicate a persistently active consumption objective. Moreover, recent purchasing activity can place consumers in an implemental mindset that increases the likelihood of follow-through (e.g., [Gollwitzer, 1990](#); [Kruglanski, Shah, et al., 2002](#); [Dhar, Huber, et al., 2007](#); [Kruglanski & Szumowska, 2020](#)).

4.1 Classifying Purchase Intention

To measure purchase intention, we classify each member-event pair using a predictive XGBoost model trained on pre-closure behavior. The model estimates whether the member would make at least one purchase in a window with the same length as the closure. We train one model for each closure duration, since the probability of at least one purchase mechanically depends on the number of days in the prediction window. Training uses relative periods -4 through -2 , validation uses the holdout pre-closure period -1 , and an additional audit evaluates control customers in period 0 , where their matched stores remain open and actual purchase behavior is observed. For treated customers in period 0 , the focal store is closed, so the outcome is counterfactual and we use the predicted probability. We classify a member-event as high intention when this predicted probability is at least 0.5 .

The features use only information available before the prediction window: recent purchase frequency, recency since the last purchase, total pre-period purchase days, store loyalty, basket composition, discount and coupon use, day-of-week purchasing patterns, demographics and push-notification history.

Table 2 reports the validation results and the most important predictors. The classifier performs similarly for treated and control customers in the holdout pre-period, with accuracy around 77%. Accuracy is also similar for control customers in period 0 , indicating that the model predicts purchase timing reasonably well in windows not used for training. The highest-ranked predictors are recent purchase intensity, total purchase history, recency and dependence on the preferred store. These patterns are consistent with the intended construct: high intention identifies customers likely to have had an active purchase goal during the closure window.

This distinction is central to our empirical design. If temporary interruptions weaken demand, the effect should be most pronounced among treated high-intention consumers, since these are the consumers whose active purchase goals are directly disrupted. In contrast, for customers without predicted purchase intention, closures represent a non-binding shock that affects exposure but does not interrupt a planned action. The empirical strategy that follows uses this distinction to separate the effect of general disruption exposure from the causal effect of blocking an active purchase. Specifically, we focus on how

Table 2: Purchase-intention classifier diagnostics

<i>Panel A: Prediction performance</i>					
Evaluation sample	<i>N</i>	Accuracy	Precision	Recall	F1
Control, during closure period 0	36,766	0.752	0.524	0.544	0.525
Control, pre-period -1	36,766	0.766	0.638	0.559	0.592
Treated, pre-period -1	8,092	0.776	0.636	0.545	0.579
Overall weighted average	81,624	0.761	0.586	0.551	0.561

<i>Panel B: Highest-ranked features</i>			
Predictor family	Mean importance	Median rank	Mean rank
Recent purchase frequency, last four weeks	28.49	1	1.23
Recent purchase frequency, last two weeks	11.54	2	7.00
Total pre-period purchase days	9.03	3	3.54
Days since last purchase	9.54	4	4.08
Preferred-store purchase share	3.38	7	7.31
Average pre-period purchase frequency	3.26	7	13.00

Notes: Panel A aggregates the duration-specific prediction audits using evaluation-sample weights. The period-0 control audit is possible because matched control stores remain open during the treated-store closure window. Panel B aggregates gain-based feature importance across duration-specific models and reports readable labels for the highest-ranked engineered features. Classifier audits are computed before final estimation restrictions.

temporary store closures affect consumer behavior along two dimensions: (i) purchase frequency and (ii) product choice, with a focus on consumers' propensity to explore new products.

4.2 Outcomes

Let i index consumers, e closure events, t time, and let L_e denote the duration in days of closure event e . For each event, we construct relative periods $t = -4, \dots, 4$, where each period is a window of length L_e and period 0 is the closure window. The main regressions omit period 0 and compare four pre-closure windows with four post-reopening windows. This construction produces a member-event-period panel in which each member-event contributes up to eight observations to the purchase-frequency analysis. We define two outcome variables on these windows.

Purchase frequency. We measure purchase intensity using purchase frequency, defined as the number of purchase days per calendar day within each event-length window. For consumer i and event e , we define purchase frequency as

$$Y_{iet}^p = \frac{1}{L_e} \sum_{t \in \mathcal{W}_e} \mathbb{1}\{\text{Purchase}_{it} = 1\}, \quad (4.1)$$

where $\mathcal{W}_e$ denotes the set of days in the relevant window pre- or post-closure and $\mathbb{1}\{\cdot\}$ is an indicator equal to one if consumer i makes at least one purchase on day t . This measure captures the extensive

margin of demand. Normalizing by closure length makes purchase frequency comparable across closure events of different durations.

Novelty-seeking. To study the composition of demand, we construct a measure of product exploration based on consumers' adoption of new products. For each consumer i and event e , we define novelty-seeking as

$$Y_{iet}^V = \frac{\sum_{t \in \mathcal{W}_e} \sum_j \mathbb{1}\{\text{Purchase}_{ijt} = 1 \text{ and } j \text{ is new for } i\}}{\sum_{t \in \mathcal{W}_e} \sum_j \mathbb{1}\{\text{Purchase}_{ijt} = 1\}}, \quad (4.2)$$

where Purchase_{ijt} indicates whether consumer i purchases product j at time t . A product is defined as new if its purchase in period t corresponds to the first time consumer i ever purchases product j in the data. This outcome measures the share of purchases devoted to new products within a given window, and therefore reflects consumers' propensity to explore the assortment. Intuitively, higher values of Y_{iet}^V indicate greater experimentation, while lower values reflect more focused or habitual consumption. By construction, novelty-seeking is only defined for consumers who make at least one purchase in the window. For consumers with no purchases, the denominator is zero and the outcome is treated as missing. This ensures that the measure captures variation in product choice conditional on participation, rather than conflating exploration with changes in purchase frequency.

These two outcomes separate the level and composition of demand. Purchase frequency captures whether consumers return to purchasing at similar rates after a disruption. Novelty-seeking captures whether they change how they allocate purchases across products.

4.3 Triple Difference-in-Differences

To quantify the causal effect of blocking a planned purchase, we estimate a triple difference-in-differences design that compares three dimensions of variation: (i) treated versus control customers, (ii) behavior before and after reopening, and (iii) high-intention versus low-intention customers.

The logic follows the canonical triple-differences design in [Gruber \(1994\)](#), who studies group-specific mandates by comparing affected and unaffected groups across states with and without policy changes before and after adoption. In that setting, the third difference removes shocks common to adopting states, shocks common to the affected group, and aggregate time effects, leaving the component specific to the treated affected group. In our setting, closure exposure is the treated-unit dimension, predicted purchase intention is the affected-group dimension, and the post-reopening period is the time dimension. Recent applications use the same idea in richer interaction settings. [Gehring and Schaudt \(2026\)](#), for example, interpret the interaction between drought exposure and insurance rollout as a triple-difference-style estimator with continuous treatment dimensions and emphasize that the main threat is an omitted factor that moves with the interaction term. This perspective motivates the coefficient decomposition and pretrend tests below.

Formally, we estimate the following pooled specification:

$$Y_{iet} = \delta^B \cdot \text{Post} \times \text{Treated} + \beta \cdot \text{Post} \times \text{Intent} + \delta^D \cdot \text{Post} \times \text{Treated} \times \text{Intent} + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}, \quad (4.3)$$

where Y_{iet} denotes the outcome of interest for member i in period t of event e . The indicator *Post* captures periods after reopening, *Treated* identifies customers exposed to a store closure, and *Intent* identifies customers predicted to have made a purchase during the closure window absent the disruption. This intent indicator is defined for both treated and control member-events. Blocking occurs only for the treated-by-intent cells. The specification includes member-closure fixed effects ϕ_{ie} , relative-period fixed effects ω_t and calendar-month fixed effects γ_m . Standard errors are clustered at the consumer level. The coefficient of primary interest is δ^D . It captures the additional post-reopening effect for treated high-intention customers. This term isolates the causal effect of having a planned purchase interrupted, over and above general post-closure shifts that affect treated customers more broadly. In contrast, δ^B accounts for the post-reopening treatment effect for low-intention customers. This coefficient reflects baseline demand shifts associated with closures, such as changes in consumer beliefs, substitution patterns or responses to reopening conditions, that are not tied to a blocked purchase. The parameter β captures differences in post-reopening behavior between high- and low-intention customers that are common across treated and control groups. This term absorbs general differences in behavior across these groups that are not driven by treatment exposure.

This specification separates the overall effect of closures into two components: a baseline demand effect (δ^B) and a blocked-intention effect (δ^D). This decomposition isolates the behavioral consequences of interrupting a planned purchase from broader changes in demand conditions.

4.3.1 Identification Assumptions and Diagnostic Tests

The identification of the blocked-intention effect relies on a set of assumptions that extend the standard difference-in-differences framework to the triple-differences setting. First, we require that parallel triple differences hold. In the absence of closures, the difference between high- and low-intention customers would have evolved similarly across treated and control groups. This condition ensures that the triple interaction term isolates the causal effect of the disruption rather than pre-existing trends.

Second, we assume that baseline demand shifts do not separately affect treated high-intention customers. That is, any post-reopening changes in prices, promotions, convenience or perceived quality affect high- and low-intention treated customers in a similar way. Under this assumption, such changes are captured by the baseline treatment effect δ^B rather than the triple interaction term.

Third, we assume that predicted-intention status and controls are valid. Intention status is determined using pre-closure behavior, ensuring that it is not endogenous to post-reopening outcomes. In addition, matched control stores provide the appropriate counterfactual for treated customers, allowing us to isolate the effect of the closure.

We assess these assumptions in the same order. First, Table 1 compares treatment and control member-events before closure. Treated and control episodes are similar in predicted purchase-intention probability and high-intention share, though treated episodes have slightly lower pre-period purchase frequency, slightly higher pre-period novelty-seeking and longer closure windows. These differences are not by themselves evidence against the design, but they motivate the fixed effects and pretrend checks below.

Second, Figure 1 verifies the first-stage shock. Purchase frequency falls sharply for treated customers during the closure window, while the control path remains available. Novelty-seeking is conditional on

purchase, so the period-0 point need not fall mechanically with access. This confirms that the closures create the intended local interruption while keeping the product-choice outcome conceptually separate from purchase incidence.

Third, we test pre-period trends using the event-study version of the triple-difference specification. For purchase frequency, the average treatment pretrend does not reject equality ($p = 0.895$), but the stricter high-intention displacement pretrend rejects ($p = 0.002$). We therefore treat the positive purchase-frequency estimate as supporting evidence that demand does not collapse after reopening, not as the cleanest causal result. For novelty-seeking, the corresponding displacement pretrend test does not reject ($p = 0.797$). The market-new novelty robustness check also passes the same test ($p = 0.515$). Appendix B reports these pretrend diagnostics.

4.4 Main Results

We begin with the regression-adjusted decomposition of the main estimates. Table 3 reports the three coefficients in equation (4.3) for purchase frequency and novelty-seeking, and then shows the implied treatment effect for high-intention episodes. The first row is the post-reopening closure effect for treated low-intention episodes. The second row captures the post-period shift for high-intention episodes that is common to treated and control customers, and is therefore not itself a treatment effect. The third row is the blocked-intention effect: the additional post-reopening change for treated high-intention episodes whose predicted purchases are interrupted by the closure.

Table 3: Decomposition of main results

Component	Purchase frequency	Novelty-seeking
Baseline closure exposure, low-intention (δ^B)	0.0009 (0.0007)	0.0313*** (0.0120)
Common high-intention post shift (β)	-0.0344*** (0.0011)	0.0381*** (0.0064)
Blocked-intention effect (δ^D)	0.0058** (0.0025)	-0.0415*** (0.0145)
Implied treatment effect for high-intention episodes ($\delta^B + \delta^D$)	0.0067	-0.0102
Observations	321,184	99,644
Within R^2	0.013	0.001

Notes: The first three rows report coefficients from the binary triple-differences specification. Standard errors, clustered at the individual level, are in parentheses. The high-intention indicator is the ex-ante predicted purchase-intention indicator. The blocked-intention effect applies to treated high-intention episodes, where the closure is predicted to interrupt a planned purchase. The final row is the arithmetic sum of the baseline closure effect and the blocked-intention effect; no separate standard error is reported for this sum. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The purchase-frequency column reveals three main patterns. The baseline treatment effect δ^B is small and statistically indistinguishable from zero. For consumers whose purchase opportunities were not binding, temporary closures do not lead to persistent changes in purchase frequency after reopening. In

other words, there is no evidence of a general demand loss driven by the disruption itself.

The coefficient β is negative and highly significant, reflecting strong mean reversion in purchase frequency among high-propensity consumers. Customers who are more likely to purchase exhibit higher pre-event activity and subsequently revert toward their typical purchase levels. This pattern is observed in both treated and control groups and therefore does not reflect a causal effect of closures.

The triple interaction coefficient δ^D is positive and statistically significant. This estimate captures the causal effect of having a planned purchase blocked by the closure. It indicates that treated high-intention customers slightly increase their purchase frequency after reopening relative to comparable high-intention customers in the control group. The magnitude is small: the estimate is 0.0058 purchase days per calendar day, or 0.58 percentage points, equal to about 11.8% of the pre-period mean purchase frequency.

These results provide no evidence that blocking a purchase weakens subsequent demand. Instead, the estimates are consistent with a modest rebound effect, whereby unfulfilled purchase intentions carry over and lead to slightly higher activity after the constraint is lifted. Because the stricter displacement pretrend diagnostics are weaker for purchase frequency than for novelty-seeking, we use this result primarily to rule out a persistent demand collapse and to motivate the analysis of product choice.

The novelty-seeking column reveals a stark contrast between average responses to closures and the behavior of consumers whose purchase intentions are directly disrupted. The baseline effect δ^B is positive and statistically different from zero. This indicates that, on average, low-intention consumers exposed to closures become more likely to try new products after reopening. This pattern is consistent with the idea that disruptions break routine consumption and increase experimentation, leading consumers to explore a broader set of products.

The coefficient β is also positive and significant, suggesting that high-propensity consumers are overall more likely to try new products. This reflects a mechanical effect: consumers who purchase more frequently are naturally more exposed to the product assortment and therefore accumulate more first-time product trials over time.

However, the key result is captured by the negative and highly significant triple interaction coefficient δ^D . This coefficient measures the causal effect of blocking a planned purchase on subsequent exploration behavior. The negative estimate implies that treated high-intention consumers become significantly less likely to try new products after reopening relative to comparable high-intention consumers in the control group. The economic magnitude is large: a 4.15 percentage point decline, equal to about 8.3% of the pre-period mean novelty-seeking rate. These findings show a clear divergence between average effects and displacement effects. While closures increase exploration on average, they reduce exploration for the subset of consumers whose purchase intentions are directly blocked. This channel stresses the importance of conditioning on whether the disruption constitutes a binding constraint.

From an economic perspective, the negative displacement effect is consistent with a shift toward more focused consumption following an interrupted purchase. Consumers who were unable to complete a planned transaction return to the market with a more targeted objective, prioritizing familiar products over experimentation. This behavior is consistent with the goal-persistence mechanism developed in Section 2 and tested in Section 5, whereby unfulfilled purchase intentions carry over and shape subse-

quent choices. The decomposition in Table 3 makes this contrast explicit: the baseline closure effect on novelty-seeking is positive, but the blocked-intention component is negative and large enough to reverse the implied treatment effect for high-intention episodes.

Overall, these empirical results show the asymmetric effects of temporary closures on demand. While store closures do not decrease purchase frequency, they substantially alter the composition of demand by reducing product exploration among affected consumers.

4.5 Robustness Checks

We further assess the robustness of our findings by replacing the binary high-intention indicator with the model’s continuous measure of ex-ante purchase-intention propensity. This approach allows us to exploit variation in the likelihood that a purchase would have been blocked for treated consumers, rather than relying on a discrete classification. The results confirm the documented patterns. The interaction between treatment exposure and the propensity score is positive and statistically significant for purchase frequency (+0.0076**), while negative and statistically significant for novelty-seeking (−0.0452**). These estimates indicate that the effect of closures becomes more positive for purchase frequency and more negative for exploration as the likelihood of a blocked purchase increases.

As an additional product-choice robustness check, we redefine novelty using market-new products: products whose first observed catalog introduction occurs in the current or immediately previous event window. This outcome is less dependent on each consumer’s own purchase history and more directly captures adoption of recently introduced items. Table 4 reports the corresponding estimates. The high-intention displacement effect remains negative and statistically significant ($\delta^D = -0.0417^{***}$, SE = 0.0115), and the continuous-score version is also negative (−0.0486***). Relative to the market-new pre-period mean of 14.6%, the binary displacement estimate implies a 28.6% decline. This robustness check is therefore consistent with the main novelty result: the reduction in exploration is not limited to products that are merely new in a consumer’s own purchase history.

Table 4: Robustness: market-new novelty

Coefficient	Estimate	SE
Baseline treatment effect	0.008	0.009
High-intention post shift	0.030***	0.005
Displacement effect	-0.042***	0.011
Continuous-score displacement slope	-0.049***	0.017

Notes: The dependent variable is market-new novelty, defined as the share of purchased products whose first observed catalog introduction occurs in the current or immediately previous event window. The first three rows use the binary high-intention DDD specification; the last row replaces the high-intention indicator with the continuous predicted purchase-intention probability. Standard errors are clustered at the individual level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5 Why Does This Happen? Goal Persistence and Consumer Type

5.1 Blocked Goals Create Focused Rebound

We propose a mechanism based on the persistence of active purchase goals. When a store closure blocks an intended purchase, the underlying goal need not disappear. Instead, the interruption may leave the goal cognitively active, increasing its salience and shaping subsequent behavior once purchasing opportunities are restored.

This idea comes from a large literature in psychology showing that unfulfilled goals remain mentally active, trigger counterfactual thinking about the missed action, and intensify the desire to complete the interrupted task (e.g., [Brehm, 1966](#); [Clee & Wicklund, 1980](#); [Fitzsimons, 2000](#); [Epstude & Roese, 2008](#); [Masicampo & Baumeister, 2011](#); [Kappes & Oettingen, 2016](#)). In our context, a consumer who is prevented from completing a purchase during a closure may return after reopening with a more focused objective, prioritizing the completion of the original purchase rather than exploring alternatives. This mechanism naturally accounts for the main empirical findings. First, treated high-intention customers do not reduce their purchase frequency after reopening. Because the underlying purchase goal persists, the missed transaction is effectively deferred rather than abandoned, leading to stable or slightly increased post-reopening activity. Second, treated high-intention customers become less likely to try new products. Because the return visit is oriented toward completing a previously intended purchase, consumers are less inclined to deviate from familiar choices and experiment with new items.

5.1.1 Implementation and Findings

A direct implication of this mechanism is that some consumers should continue to pursue their purchase goal even during the closure period. In particular, if the goal remains active, consumers may substitute toward other stores of the same chain in order to complete the intended purchase. In the data, we observe that a non-trivial fraction of treated customers make purchases at alternative stores during the closure window. Specifically, 519 treated member-events involve at least one same-chain purchase at a different store while the focal store is closed, corresponding to 7.0% of treated member-events.

To assess whether such substitution drives the results, we re-estimate the triple difference-in-differences specification after excluding all treated member-events with any during-closure purchases at alternative stores within the chain. This restriction isolates consumers whose purchase opportunities are fully blocked. The results remain qualitatively unchanged. For purchase frequency, the estimated displacement effect remains positive ($\delta^D = +0.0052^{**}$), while for novelty-seeking it remains negative ($\delta^D = -0.0323^{**}$). These estimates are slightly attenuated relative to the baseline, but preserve the same sign and statistical significance. This evidence supports the interpretation that the observed patterns are not driven by substitution across stores, but rather reflect the persistence of interrupted purchase goals. Even when substitution opportunities are removed, consumers whose purchases are interrupted exhibit a rebound in activity and a reduction in exploration, consistent with a focused return to complete the original goal.

5.2 Heterogeneity by Pre-Period Novelty

The goal-persistence mechanism also predicts that the displacement effect should differ across consumer types. Consumers who were already less exploratory before the closure have a more familiar consumption path to complete when a purchase intention is interrupted. For these consumers, counterfactual thinking about the forgone purchase should make familiar products especially salient after reopening. In contrast, consumers with high pre-period novelty-seeking have a more exploratory routine, so blocking a purchase should reduce novelty-seeking less strongly. The model therefore predicts that blocked intentions widen the novelty gap between high- and low-novelty consumers.

To test this prediction, we split member-closure episodes by pre-period novelty-seeking. For each episode, we compute the mean novelty-seeking rate over the four pre-closure windows in which the member makes a purchase. We then define $H_{ie} = 1$ if this pre-period novelty mean is above the sample median, and $H_{ie} = 0$ otherwise. The heterogeneity regression augments the main novelty-seeking specification as follows:

$$\begin{aligned} Y_{iet}^V = & \delta_L^B \cdot \text{Post} \times \text{Treated} + \beta_L \cdot \text{Post} \times \text{Intent} + \delta_L^D \cdot \text{Post} \times \text{Treated} \times \text{Intent} \\ & + \theta^B \cdot \text{Post} \times \text{Treated} \times H_{ie} + \theta^\beta \cdot \text{Post} \times \text{Intent} \times H_{ie} + \theta^D \cdot \text{Post} \times \text{Treated} \times \text{Intent} \times H_{ie} \quad (5.1) \\ & + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}. \end{aligned}$$

The omitted subgroup is low pre-novelty episodes. Thus δ_L^D is the blocked-intention effect for low pre-novelty consumers, and θ^D is the incremental blocked-intention effect for high pre-novelty consumers. The implied high pre-novelty effect is $\delta_L^D + \theta^D$. As in the main specification, member-closure fixed effects absorb time-invariant differences across episodes, while relative-period and calendar-month fixed effects absorb common time shocks. Standard errors are clustered at the member level.

Table 5 reports the two coefficients needed to interpret this heterogeneity and their implied sum. The first row shows that, for low pre-novelty consumers, a blocked purchase intention reduces subsequent novelty-seeking by 26.3 percentage points. The second row shows that the same blocked-intention effect is 49.2 percentage points higher for high pre-novelty consumers. Adding the two coefficients gives an implied high pre-novelty effect of 23.0 percentage points. The sign reversal means that the overall negative novelty result is concentrated among consumers whose pre-disruption behavior was already less exploratory. This is the pattern predicted by goal persistence: when the interruption makes an unfinished purchase goal salient, consumers with a more familiar baseline consumption path have the strongest reason to return to that path rather than explore.

This heterogeneity evidence should be interpreted as mechanism-consistent rather than as a separate randomized test. Pre-period novelty types differ in baseline behavior by construction, and may also differ in purchase frequency or other dimensions of engagement. The table therefore does not prove that pre-period novelty type itself causes the response. It shows that the main novelty-seeking decline appears exactly where the goal-persistence mechanism predicts it should appear: among consumers for whom familiar products were already the dominant pre-closure consumption path.

Table 5: Heterogeneity by pre-period novelty-seeking

Parameter or effect	Estimate	SE
Low pre-novelty effect (δ_L^D)	-0.263***	0.017
High pre-novelty increment (θ^D)	0.492***	0.024
High pre-novelty effect ($\delta_L^D + \theta^D$)	0.230	
Observations	95,986	
Member-closure fixed effects	Yes	
Relative-period and calendar-month fixed effects	Yes	

Notes: The dependent variable is novelty-seeking. Low and high pre-novelty groups are split at the sample median of episode-level pre-period novelty among members with observed pre-period novelty. The low pre-novelty row estimates δ_L^D in equation (5.1). The high pre-novelty increment estimates θ^D . The implied high pre-novelty effect is $\delta_L^D + \theta^D$; a separate standard error for this linear combination is not reported. Standard errors are clustered at the member level where reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6 Conclusion

This paper studies how consumers respond when their purchase intentions are temporarily blocked. Using panel data from a large-scale digital coffee retail platform, we exploit store closures as exogenous disruptions to purchasing opportunities and combine this variation with a triple-differences design to isolate heterogeneous responses across consumers. This approach allows us to move beyond average treatment effects and focus on how supply shocks interact with underlying consumer intent. Our main finding is that the impact of supply disruptions is fundamentally heterogeneous. While overall demand recovers quickly after reopening, suggesting that temporary constraints do not generate persistent reductions in consumption, the composition of demand changes in systematic ways. In particular, we show that treated high-intention consumers exhibit a markedly different behavioral adjustment: they become less likely to explore new products after the disruption. This negative displacement effect contrasts with average patterns in the data and highlights the importance of conditioning on consumer intent to understand the full impact of supply shocks.

We interpret these findings through a goal-based mechanism. When an active purchase intention is interrupted, the underlying goal does not disappear. Instead, it remains cognitively active and can intensify. Upon reopening, this leads consumers to prioritize goal completion, directing their purchases toward familiar or previously chosen products. The heterogeneity analysis supports this interpretation: the reduction in exploration is concentrated among consumers with lower pre-period novelty-seeking, for whom the familiar purchase path is likely to be more salient. The evidence is therefore consistent with counterfactual thinking and goal persistence as a behavioral explanation for the observed reduction in novelty seeking among high-intention consumers, despite the absence of long-run effects on overall purchase frequency.

Our findings contribute to the marketing literature in several ways. First, we provide empirical evidence on the role of goal persistence and state dependence in shaping consumer behavior under temporary constraints. While prior work has emphasized habit formation and inertia, our results highlight the

importance of actively maintained purchase goals and their behavioral consequences when disrupted. Second, we show that supply-side shocks can have distinct effects on the level and composition of demand, and that these effects depend critically on consumer heterogeneity. By focusing on consumers for whom the closure likely binds, we uncover behavioral responses that are masked in aggregate analyses. Furthermore, our study contributes to a broader understanding of product discovery by showing how temporary interruptions can have lasting effects on consumer choice, even when demand levels recover quickly.

From a managerial perspective, our results suggest that demand recovery following disruptions may be incomplete in important dimensions. Although aggregate sales may rebound, firms should be aware that certain segments of consumers, particularly those with high purchase intent and low baseline novelty-seeking, may disproportionately revert to familiar products. This can limit the effectiveness of new product introductions and reduce the overall rate of product discovery. To address this, firms may benefit from designing targeted interventions that help returning consumers consider alternatives rather than simply complete the interrupted purchase path. Overall, our study shows that understanding demand dynamics requires jointly considering supply constraints, consumer goals and consumer heterogeneity. Temporary disruptions do not simply shift demand over time, they reshape how consumers make choices once constraints are lifted. By highlighting the role of consumer heterogeneity and behavioral mechanisms, our results provide a more nuanced view of how demand adjusts to shocks and offer new insights into the interplay between supply-side conditions and consumer decision-making.

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Part

Online Appendix

Table of Contents

Table of Contents	i
A Data Construction and Measurement	ii
A.1 Closure Registry and Member-Event Sample	ii
A.2 Outcome Timing and Variable Definitions	iii
A.3 Purchase-Intention Classifier	iv
B Identification Diagnostics	v
B.1 Pretrend Tests	v

A Data Construction and Measurement

This appendix records the construction details that are needed to interpret the empirical design but are too mechanical for the main text. The unit of analysis is a member-closure event. A member can enter the analysis only through a closure event for which her pre-closure purchase history identifies a meaningful focal store.

A.1 Closure Registry and Member-Event Sample

The closure registry starts from a store-day panel. A candidate closure is a consecutive spell in which a store records no transactions. The raw detection step identifies 101 zero-demand spells. Because many university and college locations have long closures that reflect campus operating calendars rather than short disruptions, the analysis excludes those locations. The maintained paper design then focuses on the 18-closure registry used for the headline estimates.

Table A.1: Sample construction for the 18-closure design

Step or object	Count	Description
Transactions	10.6 million	Raw purchase records from the coffee-chain transaction data.
Customers	779,000	Customers observed in the transaction data.
Stores	260	Physical stores in the operating network.
Raw zero-demand closure spells	101	Consecutive store-day spells with no transactions.
University or college closure spells excluded	60	Long campus-related closures removed from the main disruption design.
Non-university spells before final registry filters	41	Raw non-university closure candidates before the treatment-control registry restrictions.
Main retained registry	18 closures	Headline registry used in the paper. Closure durations range from 10 to 29 days.
Member-closure episodes	40,148	Final episode-level sample underlying the purchase-frequency panel.
Treated episodes	7,390	Members whose pre-closure preferred store is the closing store.
Control episodes	32,758	Members whose pre-closure preferred store is a matched open control store.
Treated high-intention episodes	2,041	Treated episodes with predicted purchase intention above the classifier threshold, equal to 27.6% of treated episodes.

Notes: The full transaction counts describe the raw data environment. The final episode counts describe the main 18-closure estimation sample. Treatment and control assignment use pre-closure preferred stores and the same minimum purchase and store-loyalty restrictions for both groups.

For each closure, treated members are customers whose preferred pre-closure store is the store that closes. Control members are customers whose preferred pre-closure store is one of the matched control

stores that remains open. Preferred stores are defined using pre-closure purchases, and the sample requires at least five pre-closure purchase days and a preferred-store purchase share of at least 80%. These restrictions make store assignment behaviorally meaningful: the closure is a relevant local disruption only if the customer had repeatedly relied on that store before the event.

A.2 Outcome Timing and Variable Definitions

Each closure event defines relative periods $t = -4, \dots, 4$. Every relative period has the same length as the closure, denoted L_e , and period 0 is the closure window itself. The main post-reopening regressions omit period 0 and compare four pre-closure windows with four post-reopening windows. This convention produces 321,184 member-event-period observations for purchase frequency. Product-choice outcomes are observed only when a member purchases in the corresponding window; the main novelty-seeking regression uses 99,644 observations after the final estimation restrictions.

Table A.2: Key variables in the empirical design

Variable	Construction	Interpretation
Treated	Preferred pre-closure store is the closed store.	Exposure to the local store closure. This is not the same as having a purchase blocked.
Control	Preferred pre-closure store is a matched store that remains open.	Counterfactual local-store path for treated customers.
Intent	Ex-ante predicted probability of at least one purchase during the closure-length window is at least 0.5.	Predicted active purchase intention during period 0. It is defined before observing post-reopening outcomes.
Blocked intention	$\text{Treated} \times \text{Intent}$.	The closure is predicted to bind: the customer is exposed to the closure and likely would have purchased during the closure window.
Post	Relative periods 1 through 4.	Post-reopening windows. Period 0 is the closure window and is omitted from the main DDD regressions.
Purchase frequency	Purchase days in the relative-period window divided by L_e .	Extensive-margin purchase intensity, normalized for closure length.
Novelty-seeking	Share of purchased products in the window that are new to the member.	Product exploration conditional on purchase. Missing when the member makes no purchase in the window.
Market-new novelty	Share of purchased products whose first catalog appearance occurs in the current or immediately previous event window.	Robustness measure focused on adoption of recently introduced products rather than products merely new to a given member.

Notes: The main text uses “high intention” for episodes with predicted purchase intention above the threshold. The word “blocked” is reserved for treated high-intention episodes, because control high-intention episodes remain able to purchase from their focal store.

A.3 Purchase-Intention Classifier

The purchase-intention classifier is used to separate closure exposure from binding interruption. The model predicts whether a member would purchase at least once in a window with the same length as the closure. This target depends mechanically on window length, so the implementation trains one XGBoost model for each observed closure duration. Training uses pre-closure relative periods -4 through -2 , validation uses the holdout pre-period -1 , and an additional audit uses control customers in period 0 because their matched focal stores remain open. The period-0 audit is a validation exercise; the intent variable used in the DDD is the ex-ante predicted score.

Table A.3: Classifier implementation details

Component	Implementation
Prediction target	Indicator for at least one purchase in a closure-length window.
Model family	XGBoost binary classifier, estimated separately by closure duration.
Training windows	Relative periods -4 , -3 and -2 .
Holdout validation	Relative period -1 , separately audited for treated and control customers.
Additional audit	Control customers in period 0 , where the matched control store remains open and actual purchase behavior is observed.
Scored sample	Treated and control member-closure episodes in the ex-ante period-0 panel.
Classification rule	High intention equals one if the predicted period-0 purchase probability is at least 0.5 .
Feature families	Recent purchase frequency, recency, total purchase history, preferred-store loyalty, basket composition, discount and coupon use, day-of-week purchasing patterns, demographics and push-notification history.

Notes: The prediction-performance table in the main text reports accuracy, precision, recall and F1 for the main validation slices. The appendix table emphasizes timing and information sets, because these are the key design concerns for interpreting the classifier as ex ante.

B Identification Diagnostics

This appendix reports the formal diagnostics behind the identification discussion in Section 4.3.1. The main design has three dimensions: treatment exposure, post-reopening timing and predicted purchase intention. The key identifying restriction is therefore a triple-difference restriction: absent the closure, the treated-control gap for high-intention consumers would have changed like the treated-control gap for low-intention consumers.

B.1 Pretrend Tests

Table B.1 reports joint tests from the event-study analog of the main DDD specification. Each test uses the three pre-reopening event-time coefficients relative to period -1 . The average treatment pretrend tests whether treated and control consumers move similarly before the closure. The baseline treatment pretrend tests the treated-control pretrend among low-intention consumers. The high-intention displacement pretrend tests the pre-period coefficients on the treated-by-intention interaction, which is the event-study analog of δ^D . The continuous-slope test replaces the binary high-intention indicator with the predicted purchase-intention score.

Table B.1: Pretrend diagnostics for event-study specifications

Outcome	Test	Restrictions	p -value
Purchase frequency	Average treated-control pretrend	3	0.895
Purchase frequency	Baseline treatment pretrend for low-intention consumers	3	0.055
Purchase frequency	High-intention displacement pretrend	3	0.002
Purchase frequency	Continuous intention-slope pretrend	3	0.007
Novelty-seeking	Average treated-control pretrend	3	0.186
Novelty-seeking	Baseline treatment pretrend for low-intention consumers	3	0.477
Novelty-seeking	High-intention displacement pretrend	3	0.797
Novelty-seeking	Continuous intention-slope pretrend	3	0.778
Market-new novelty	Average treated-control pretrend	3	0.177
Market-new novelty	Baseline treatment pretrend for low-intention consumers	3	0.990
Market-new novelty	High-intention displacement pretrend	3	0.515
Market-new novelty	Continuous intention-slope pretrend	3	0.721

Notes: The table reports joint Wald tests of pre-reopening coefficients in event-study specifications. Purchase frequency uses 321,184 member-event-period observations. Novelty-seeking and market-new novelty use 99,644 observations because product-choice outcomes are defined only in windows with purchases. Standard errors are clustered at the individual level.

The pretrend diagnostics support the main novelty interpretation more strongly than the purchase-frequency interpretation. For purchase frequency, the average treated-control pretrend is flat, but the

high-intention displacement pretrend rejects equality. This is why the purchase-frequency result is interpreted cautiously as evidence against a demand collapse rather than as the paper's cleanest causal estimate. For novelty-seeking, the high-intention displacement pretrend does not reject, and the same is true when the high-intention indicator is replaced by the continuous purchase-intention score. The market-new novelty robustness check shows a similar pattern.